

## CSC 487 Practice Quiz 1

1. Find a solution (by hand) for the following binary linear classification problem:

positive class: (1,1), (1,-1)

negative class: (-1,1), (-1,-1)

(Write down the parameters of your linear classifier solution.)

Verify that your solution is correct by calculating the prediction of the classifier for each point.

### Solution:

There are many (infinite!) solutions, but a simple answer is:  $\mathbf{w}=(1,0)$  and  $b=0$ .

To verify, we plug each point into  $z = \mathbf{w} \cdot \mathbf{x} + b$  and check that the sign of  $z$  matches the class label.

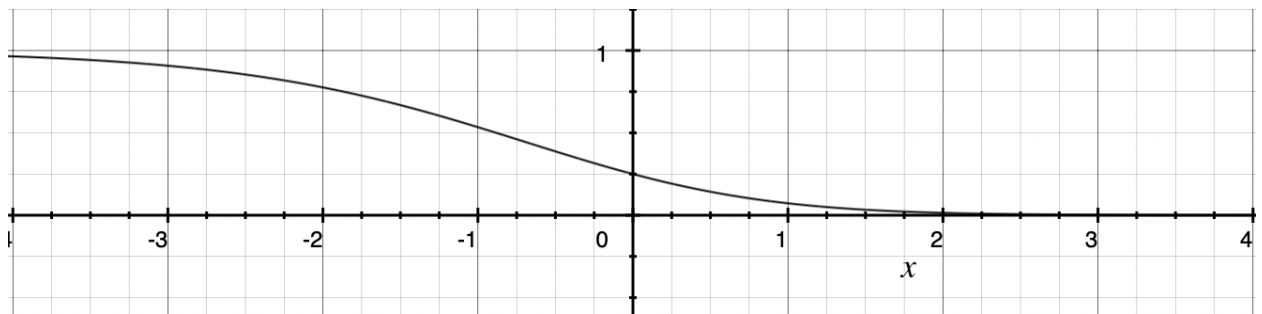
Positive points:  $1 \cdot 1 + 0 \cdot 1 + 0 = 1 > 0$  and  $1 \cdot 1 + 0 \cdot (-1) + 0 = 1 > 0$

Negative points:  $1 \cdot (-1) + 0 \cdot 1 + 0 = -1 < 0$  and  $1 \cdot (-1) + 0 \cdot (-1) + 0 = -1 < 0$

2. This is the “Savage loss” function for a positive point in a binary linear classifier:

$$L(\mathbf{x}; \mathbf{w}) = \frac{1}{(1 + e^z)^2}$$

Here’s a graph of it:



What would be the gradient descent update rule for this loss function?

**Solution:**

The gradient descent update rule is

$$\mathbf{w}' = \mathbf{w} - s \nabla_{\mathbf{w}}$$

By the chain rule, we know that

$$\frac{\partial L}{\partial w_j} = \frac{\partial L}{\partial z} \frac{\partial z}{\partial w_j}$$

Using power rule and chain rule we have

$$\frac{\partial L}{\partial z} = \frac{-2e^z}{(1 + e^z)^3}$$

as shown in class we have

$$\frac{\partial z}{\partial w_j} = x_j$$

Putting them together we have

$$\frac{\partial L}{\partial w_j} = \frac{-2e^z}{(1 + e^z)^3} x_j$$

and so

$$\nabla_{\mathbf{w}} = \frac{-2e^z}{(1 + e^z)^3} \mathbf{x}$$

Now we can plug this into the gradient descent update rule to get

$$\mathbf{w}' = \mathbf{w} + s \frac{2e^z}{(1 + e^z)^3} \mathbf{x}$$

*Extra challenges:*

Write the derivative in terms of the Savage loss itself and the sigmoid function:

$$\sigma(x) = \frac{e^x}{1 + e^x} = \frac{1}{1 + e^{-x}}$$

**Solution:**

$$\mathbf{w}' = \mathbf{w} + s 2\sigma(z)L(\mathbf{x}; \mathbf{w})\mathbf{x}$$

Interpret the design of the Savage loss function and how the update rule works.

**Solution:**

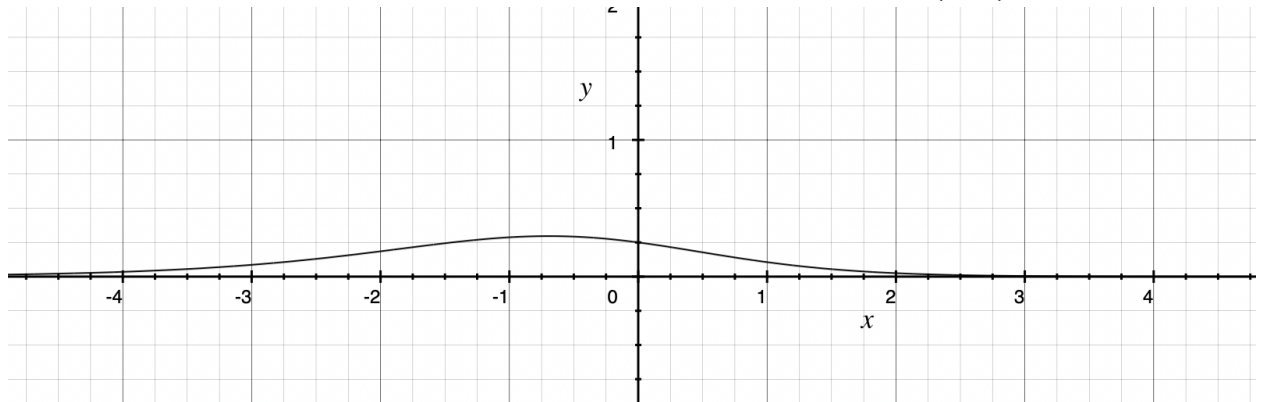
Our goal in gradient descent is to minimize the loss function, and we are assuming that  $\mathbf{x}$  has a positive class label. Looking at the loss function curve, you can see that it is minimized as  $z$  becomes a large positive value, and it increases as  $z$  becomes negative. So, minimizing the loss function will ensure that  $z$  has the correct sign.

Another interesting aspect of the loss function is that it is bounded between 0 and 1 and saturates on either side. So, very wrong values of  $z$  (very negative values) are penalized about the same as somewhat wrong values of  $z$ . This reflects the “outlier-robust” nature of the Savage loss, which means that it is not greatly affected by outlier points that aren’t able to be correctly classified.

To understand the update rule, we can plug our update rule into the formula for  $z$ :

$$\begin{aligned} z' &= \mathbf{w}'^T \mathbf{x} \\ &= (\mathbf{w} + s \frac{2e^z}{(1+e^z)^3} \mathbf{x})^T \mathbf{x} \\ &= \mathbf{w}\mathbf{x} + s \frac{2e^z}{(1+e^z)^3} \mathbf{x}^T \mathbf{x} \\ &= z + s \frac{2e^z}{(1+e^z)^3} \|\mathbf{x}\|^2 \end{aligned}$$

Recall from class that  $\|\mathbf{x}\|^2$  is always positive. Here’s a graph of the  $\frac{2e^z}{(1+e^z)^3}$ :



You can see that it is also always positive. So, the new  $z$  value will always be higher than the old value, which makes sense for a positive-labeled point.

However, you can also see that the graph flattens out on either side. That makes sense for positive points, since they are already correctly classified and so the weight doesn’t need to change. But also, for outlier points where the  $z$  value is very wrong (very negative), the loss function essentially “gives up” and stops trying to push on that point.